Electrophoresis Power Supply

(Model: DYY－15D / Cat.#:113 - 1540)

User's Manual

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Special Statement

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Warranty & Liability

Our company guarantees that this unit deliveredhas been thoroughly tested and inspected to ensure that it meets the specifications mentioned in this manual, and is warranted against defective material and workmanship for a period of 18 months from the date of shipment ex-factory.

Our company will repair or replace the defective equipment returned during the warranty period free if the equipment has been used under normal laboratory conditions and in accordance with the instruction in this manual, we shall in no event be liable for the defects including 1.) Damage caused by accident, accidental misuse or abuse; 2.) Damage caused by disaster; 3.) Repair or modification by anyone else without our authorization; 4.) Defects caused by improper operation; 5.) Use of cables or connectors not specified for this unit.

A return authorization must be obtained from us before returning any product for repair on a freight prepaid basis.

Our company reserves the right to alter the specification or improve any thing of its products without any prior notice, which will enable us to implement the developments in time when they arise.

For any inquiry or request for repair service,contact us,also please provide the model or serial number and purchase date of your instrument.

Special suggestion

Thanks for choosing DYY–15D power supply. To insure the best performance from our electrophoresis power supply, please become fully acquainted with these operating instructions before using it. We recommend that you should first read these instructions carefully. This User's Manual should be with the power supply for your convenience, once the manual is lost, please ask for it from us.

This instrument is for indoor laboratory use only. Please don't use it for purposes other than those for which it is intended.

Introduction of the power supply

The electrophoresis power supply is used while electrophoresis technology is applied. It is used in clinical test in hospitals, in the detection in polices and politics and law organs. It is also used in the laboratories of universities and colleges.

DYY － 15D power supply is controlled by micro-computer, LCD can show all the parameters including voltage, current, power, time, mode, running status, etc.

DYY － 15D power supply is one of our multi -purpose power supplies, providing constant voltage, constant current and constant power for electrophoresis applications. DYY － 15D power supply is ideal for DNA/RNA separations, Protein, Agarose, DNA sequencing, Isoelectric focusing, Westing blotting, Semi-dry blotting, 2-dimensional electrophoresis ect.

Constant voltage, constant power or constant current running with automatic crossover.

DYY － 15D power supply provides four operating modes: standard, time, volt-hour and step. When you operate in any one of the four modes, you simply enter the desired power conditions and begin the running.

DYY － 15D power supply is programmable, with the capacity to store up to 9 different programs, each with 9 steps. The programs are stored in the power

supply, and you can recall them at any time, which is especially useful when you select step mode.

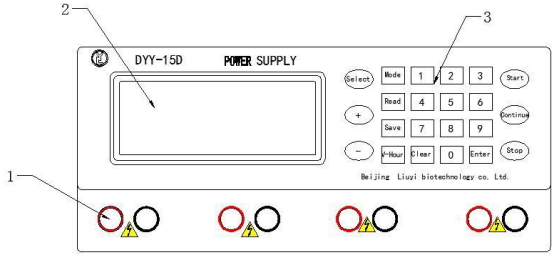
The parameters including voltage, current, power and time can be adjusted finely during the running.

Automatic detection of no_load, over_voltage、 short circuit, over_load, sudden_load change, leaks electricity, etc.

Four groups of output terminals are for running four units of electrophoresis cell simultaneously.

Figure 1: Front panel

On the front panel, there are a LCD screen, membrane numeric keypad, ON/OFF switch and output terminals for four units of electrophoresis cells.



(Figure 1)

1.) Output terminals: The leads are color coordinated to the output terminals (positive / red; negative / black)

2.)LCD screen: To display all the parameters including voltage, current, power, time, mode, running status, etc.

3.)Keyboard: Keyboard consists of function keys, numeric keys and control keys, function keys includes "mode" key, "read" key, "save" key, ect., numeric keys include from "0" to "9" key and control keys include "select" key, "+" key, "-" key, etc.

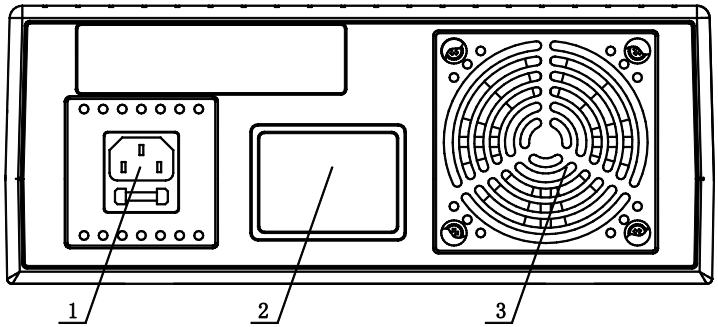
Key	Description
"Select"	Selects the parameters when you need to set or adjust them.
"+"	Changes the value of the selected parameter.

"-"	Or adjust the value of "s" in the step mode.
"Mode"	Selects the running mode including Standard mode, time mode, step mode and V-Hour mode.
"Read"	Reads out the programs stored in M0 – M9.
"Save"	Stores the programs into M0 – M9.
"V-hour"	Starts the V-hour mode directly in the step mode.
"Clear"	Deletes the value of parameters when you need to reset or adjust them.
"Enter"	Confirms each the step of the operation.
"Start"	Starts the running.
"Continue"	Continues the running after the pause.
"Stop"	Terminates the output of the running and displays the final run parameters.

4.) ON/OFF switch: ON/OFF main power switch.

Figure 1: Rear panel

The rear panel is shown in Figure 2.



Power cord receptacle.

1) Fuse socket: Unscrew it in counterclockwise direction, you can open it and can change the fuse if necessary; screw it in clockwise direction, you can firm it after you change the fuse.
2) Cooling fan: when the output current reach the set value, the fan will start working.

(* If the power supply is suitable for the requirement of 110V, here, ~220V 50HZ will be changed to ~110V 50 – 60Hz.)

Specification

1.) Power supply: A.C 220V ± 10% (50Hz ± 2%)
2.) Input power: Max. 600VA
3.) Output voltage: (10 ~ 5000)V (fully adjustable in 1V / step)
4.) Output electric current: (4 ~ 300) mA (fully adjustable in 1mA / step)
5.) Output power: (4 ~ 300)W (fully adjustable in 1W / step)
6.) Four groups output terminals in parallel
7.) Time control: 1 minute to 99 hours and 59 minutes
8.) Ripple coefficient: <2%
9.) Stability: constant voltage ≤1% constant electric current: ≤2% constant power ≤3%
10.) Set point accuracy: constant voltage ≤2% constant electric current: ≤3% constant power ≤5%
11.) Nine programs each with nine steps
12.) Ambient temperature: (5~40) ℃/ (41~104) °F
13.) Relative humidity: ≤80%
14.) Dimension: (278×349×110) mm
15.) N.W.: about 4.0 kg

Safety information

The DYY-15D power supply produces high voltage outputs that are electrically isolated from earth ground through a protective impedance to minimize the risk of electrical shock to the user.

For proper ventilation, be sure that there is at least 5 – 10 (cm) of space behind the apparatus, and at least 5 (cm) of space around the power supply. Don't block the fan at the rear panel.

Connect the apparatus to a three- prong, grounded AC power outlet, using the power cord provided with the power supply. Don't cut and splice the power cord. When you pull the power cord from the wall outlet, please make sure to hold the plug attachment and not pull the

power cord directly, and please don't hold the plug by wet hand.

Do not operate the apparatus in extreme humidity (above 80%), and avoid condensation, which can short the internal electrical circuits of the power supply, by letting the power supply equilibrate to ambient temperature when taking it from a colder to a warmer environment.

When you carry out the experiment, please don't put the electrophoresis cell on the top of the electrophoresis power supply lest the liquid will spill on or into the power supply. You'd better keep a safe distance between the electrophoresis cell and the power supply.

Please don't touch the buffer solution or the sample in the electrophoresis cell for the high output of the voltage. You'd better turn off the power supply and disconnect the electrophoresis cell from the power supply lest danger occurs.

Keep the apparatus as dry and clean as possible. You can wipe it regularly with a soft, wetted cloth. And be sure that the apparatus is dry, and not wetted before you use it.

The molded leads supplied with the electrophoresis cell are inserted into the power supply's high voltage output terminals, and connect the output terminals only to an electrophoresis cell.

Take care so that the power supply is not dropped to avoid damaging the apparatus or injuring yourself.

Never place the power supply near or over a radiator or stove.

Avoid connecting a high voltage output lead to earth ground.

To avoid any risk of injury, the instrument should only be operated by properly trained personnel.

Please contact us if you need the help, and never open the apparatus by yourself.

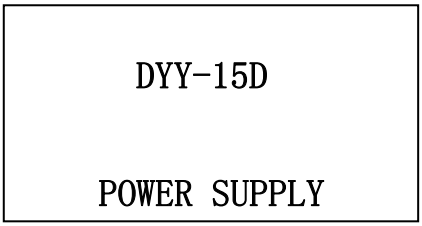
Operating instructions

Part 1: Description of the LCD

1. Place the DYY －15D power supply on a level table, connect the power cord to the power cord receptacle at the rear panel and to a suitable grounded three-prong AC power outlet.

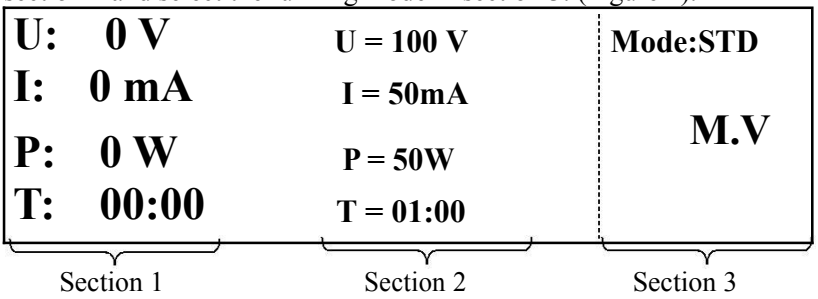
   Note: We recommend you to use only one hand when making or breaking a connection to avoid making a complete circuit across your body, and make sure that your other hand is not touching anything that grounds you.

2. Connect the electrophoresis cell to the electrophoresis power supply with the leads that are color coordinated to the output terminals (About the operation steps of the electrophoresis cell, please read the relevant user's manual for the details).
   Note: If you want to run two or more units of electrophoresis cells at the same time, we suggest that you select constant voltage as the running condition because the working value of the current shown in the LCD screen is the sum of the currents of the different electrophoresis cells, but the voltages of different electrophoresis cell are the same.
3. To turn on the "ON/OFF" switch on the front panel;
4. The LCD will show the following information.



(Figure 3)

After 2 seconds, the electrophoresis power supply will beep four times, and LCD screen will show you the following user-interface. Then, you can set the parameters that you need in section 2 and select the running mode in section 3. (Figure 4).



(Figure 4)

The user-interface consists of three section:
Section 1: large U, I, P, T are actual running values;
Section 2: small U, I, P, T are pre-assigned values (all the values

defaults after you turn on the "ON/OFF" switch);

U = 100 V
I = 50 mA
P = 50 W
T = 01:00
(VH = 1000)

Section 3: Area for displaying the information of the running status, the running mode, etc. like mode: STD, TIME, VH, STEP.

M.V or H.V in section 3 means the setting status of the output.

The scope for setting H.V: U: (1500 ~ 5000)V I: (4 ~100) mA P: (4 ~ 300)W (If the value of the voltage in section 2 is more than 1500V, after you start the running, LCD will display H.V);

The scope for setting M.V: U: (10 ~ 2000)V I: (4 ~300) mA P: (4 ~ 300)W (If the value of the voltage in section 2 is less than or equal to 1500V, after you start the running, LCD will display M.V)

Part 2: The methods of starting up the running programs

There are three methods to start up the running programs:

① You can edit the new running program, please see the "Part 4" for the details;

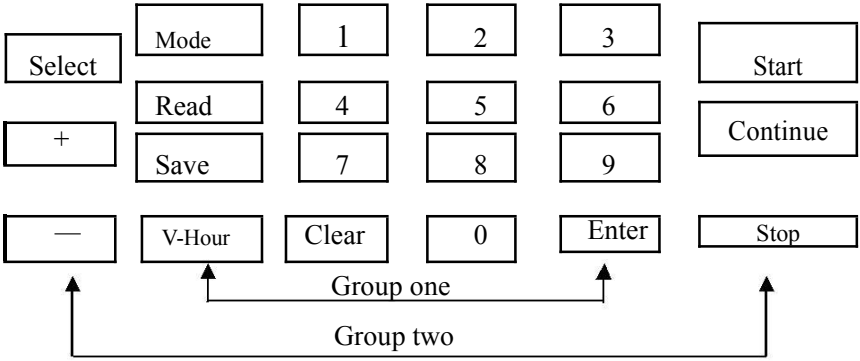
② Press "Read" key to read out the running program of last time stored in M0, then, press "Enter" key for the confirmation;

③ Press "Read" key, "n" key and then "Enter" key to read out the running program stored in Mn (n = 1 - 9).

Attention: You should check the program carefully before you press the "Start" key to start the running.

Part 3: Introduction of the keyboard

1.) The position of different keys on the keyboard (Please refer to Figure 5)



(Figure 5)

2.) The functions of different keys:

The 16 pieces of small keys in the middle part (group one) are for setting working mode and the value of the parameters. The 6 pieces of bigger keys on right and left side (group two) are for confirming and changing the output status.

Part 4: Operation of the power supply

1. Standard Mode Operation: The standard mode is used for the most basic electrophoresis experiments. The power supply is programmed to run in constant voltage, constant current or constant power within the set limits.

Here, there is an example to show you how to set the parameters in the standard mode.

Example 1: If you want to work under the following running conditions: Constant voltage, U = 1,000 V; I ≤ 100 mA; W ≤ 100W; The period of running is 3 hours and 20 minutes. And the electrophoresis power supply will stop the output automatically when all the time has elapsed. The operation steps are as follows:

⑴ Connect the electrophoresis cell to the electrophoresis power supply with the leads that are color coordinated to the output terminals (About the operation steps of the electrophoresis cell, please read the relevant user's manual for the details);

⑵ Turn on the electrophoresis power supply by pressing the

"ON/OFF" button, then LCD will show the salutatory and the electrophoresis power supply will beep four times at the times. Refer to figure 3.

(3) The default mode of the power supply is the standard mode. Here, on the top right corner of LCD will show "Mode: STD".

Attention: The "Mode" key is for selecting running mode. If you press "Mode" key, the standard mode will be changed to other modes. The order of the running mode will change as follows, if you press "Mode" key continually:

→ STD → Time → VH → Step

Standard mode: Output will not be stopped when all the set time has elapsed;

Time mode: Output will be stopped when all the set time has elapsed;

VH mode: Output will be stopped when the product of the output of voltage and the running time reaches the set value;

Step mode: Output can be carried out according to the given steps (1 ~ 9) and running mode (time mode or VH mode).

(4) Use the following procedure to set voltage, current, power and running time.

① Press numeric keys to input the voltage directly, here, we press "1000" to set the voltage equaling to 1000 V.

② Press "Select" key to select the current – "I", then, we press "100" to set the current equaling to 100 mA.

③ Press "Select" key to select the power – "P", and then, we press "100" to set the current equaling to 100W.

④ Press "Select" key to select the Timing – "T", press "320" to set the timing equaling to 03:20 (three hours and twenty minutes).

Note: 1.) If you find that you input the wrong number, you can press "Clear" key to erase the wrong number and then input the right ones.

2.) The set limits of U,I and P are as follows:

U: (10 – 5000) V;
I: (4 - 300) mA;
P: (4 - 300) W.

If the parameter that you input is not within the set limits, after you

press "Start" key, LCD will show you "U (I or P)_Data?!" in the middle section on the right area. Then, you can input the right one directly.

3.) If the voltage that you input is more than 1500V, the current that you will input must be less than or equal to 300mA, otherwise, LCD will show you "U_? I _?! " in the middle section on the right area after you press "Start" key.

⑸ Press "Start" key to begin the running. The power supply will beep two times to remind you of high output voltage. And at the same

time, LCD will show the glittering " ϟϟ " and the actual running time in section 3. (Figure 4)

Note: 1.) The procedure is saved into M0 after you press the "Start" key to begin the running;

2.) If the pre-assigned values in section 2 (Figure 4) are right, "U:" on the top left corner of LCD will be glittering to show that the voltage is constant, and the actual running value of current (I) and power (P) should be less than the pre-assigned values;

3.1) If it is "I:" or "P:" that is glittering, it means that "I:" or "P:" has reached the pre-assigned values, which limits "U:". The reason may be that the resistance value of the sample is low, please check the sample. If the sample is OK, then the pre-assigned values are wrong and should be changed;

3.2) You can change U, I, P by pressing "+" key or "-" key directly: "+" key: After you select the parameter, press this key at a time, the value of the parameter will increase 1, and if you press the key without a break, the value will increase fast and continuously, and relax the key
when the parameter is changed to what you need;

"-" key: After you select the parameter, press this key at a time, the value of the parameter will decrease 1, and if you press the key without a break, the value will decrease fast and continuously, and relax the key when the parameter is changed to what you need;

⑹ If you want to stop the running, press "Stop" key, LCD will display "Stop" and the running time in section 3 (Figure 4). And the power supply will beep three times to remind you that the output has been shut down;

⑺ If you want to continue the running from the elapsed time, press "Continue" key, LCD will display "Continue!" in section 3 (Figure 4), and the power supply will beep two times to remind you of high output

voltage, and then, LCD will show the glittering " ϟϟ " in section 3. (Figure 4). The running time starts from the elapsed time.

⑻ If you want to continue the running from the beginning (00:00:00), press "Start" key, LCD will display "Start!" in section 3 (Figure 4), and the power supply will beep two times to remind you of

high output voltage, and then, LCD will show the glittering " ϟϟ " in section 3 (Figure 4). The running time starts from the beginning (00:00:00), again.

⑼ When all the pre-assigned time has elapsed, the power supply will beep 13 times to remind you, but the output will not stop until you press "Stop" key.

Note: 1.) After the power supply begins the running, only one of them (U, I ,P) can keep constant, depending on the pre-assigned values and the load of the power supply.

2.) Set the parameter (voltage, current or power) equaling to the value that you want it constant to, and then input the max. values that the power supply allows of the other two.

2. Time Mode Operation: DYY － 15D power supply can work in time mode for a set period of time. And in the time mode, when all the pre-assigned time has elapsed, the power supply will stop the output and beep continuously to remind you until you press any key to stop the alarm. LCD will show "End:" and the running time in section 3 (Figure).

Press "mode" key to change the running mode from "STD" to "Time", and then you can input the pre-assigned values of each parameters, and other operation steps are the same as those of the operations in standard mode.

3. Volt-Hour Mode Operation: Some electrophoresis experiments need the power supply continuously integrates the voltage as a function of time and reports this value in volt-hours. For these experiment, you can select volt-hour mode.

The value of the Volt-Hour equals to the product of the output of voltage and the running time. For example, if the output voltage (V) is 1000V, and the running time (T) is 2 hours, then, the value of the Volt-Hour is 2000. During the running, if the actual output voltage is only 500V, the power supply will adjust the time from 2 hours to 4 hours

automatically. And the operations of inputting voltage (V), current (I) and power (P) are the same as those of operations in the standard mode.

Example 2 : VH = 2000, U = 1000V, I = 100mA and P = 100W, the operation steps are as follows:

(1) Connect the electrophoresis cell to the electrophoresis power supply with the leads that are color coordinated to the output terminals (About the operation steps of the electrophoresis cell, please read the relevant user's manual for the details);

(2) Turn on the electrophoresis power supply by pressing the "ON/OFF" button, then LCD will show the salutatory and the electrophoresis power supply will beep four times at the times. Refer to figure 3.

(3) The default mode of the power supply is the standard mode. Here, on the top right corner of LCD will show "Mode: STD".

(4) Press "Mode" key to change the running mode from "STD" to "VH";

(5) Press numeric keys to input the voltage directly, here, we press "1000" to set the voltage equaling to 1000 V.

(6) Press "Select" key to select the current – "I", then, we press "100" to set the current equaling to 100 mA.

(7) Press "Select" key to select the power – "P", then, we press "100" to set the power equaling to 100 W.

(8) Press "Select" key to select the volt-hour – "VH", then, we press "2000" to set the VH equaling to 2000.

(9) Press "Start" key to begin the output, the power supply will beeps two time to remind you of the high output voltage. And at the same time, LCD will show the glittering " ϟϟ " and the actual running time in section 3. (Figure 4).

When the product of the product of the output of voltage and the running time reaches 2000, the power supply will stop the output automatically and keeps beeping to remind you until you press any key to stop the alarm.

4. Step Mode Operation: The step mode allows up to 9 different procedures with 9 steps of running conditions to be programmed and run in sequence. During the running, the power supply will automatically

switch from one procedure to another. And the time mode or the volt-hour mode can be used in the step mode. The step mode is ideal for the electrophoresis experiments that require continuous output adjustments like discontinuous SDS-PAGE and Isoelectric focusing.

Example 3: There are 3 steps in one experiment, and the parameters of each step are as follows:

Step 1 (S = 1) : Constant current I = 45mA, U = 1500V, P = 70W and T = 00:10 (10 minutes);

Step 2 (S = 2) : VH mode U = 500V, I = 100mA, P = 50W and VH = 150;

Step 3 (S = 3) : Constant power P = 30W, U = 1000V, I = 80mA, T = 00:15 (15 minutes).

The operation steps are as follows:

⑴ Connect the electrophoresis cell to the electrophoresis power supply with the leads that are color coordinated to the output terminals (About the operation steps of the electrophoresis cell, please read the relevant user's manual for the details);

⑵ Turn on the electrophoresis power supply by pressing the "ON/OFF" button, then LCD will show the salutatory and the electrophoresis power supply will beep four times at the times. Refer to figure 3.

⑶ The default mode of the power supply is the standard mode. Here, on the top right corner of LCD will show "Mode: STD".

⑷ Press "Mode" key to change the running mode from "STD" to "Step";

⑸ Press "3" directly to set SN = 3; **Note:**

The value of SN is 1 to 9.

⑹ Press "Select" key to select the voltage – "V", then, we press "1500" to set the voltage equaling to 1500 V.

⑺ In the same way, set I = 45mA, P = 70W and T = 00:10. The first step is completed.

⑻ Press "+" key to change the value of S from "S = 1" to "S = 2".

⑼ Press "500" directly to set the voltage equaling to 500 V.

⑽ Press "Select" key to select the current – "I", then, we press "100" to set the current equaling to 100 V.

⑾ In the same way, set P = 50W。

⑿ Press "Select" key to select the timing – "T", then, press

"V-Hour" key, "T" will change to "VH".

⑬ Press "150" to set VH equaling to 150. The second step is completed.

⑭ Press "+" key to change the value of S from "S = 2" to "S = 3". ⑮ In the same way, set U = 1000V，I = 80mA, P = 30W and T = 00:15. The third step is completed.

⑯ Press "+" key, return to "S = 1".

⑰ Press "Start" key to begin the running.

Note: 1.) Regardless of the value of SN, only when "S = 1" after you finish the setting of all the steps, can you start the running.

2.) During the running, when the power supply switches from one step to next step, it will beep 5 times to remind you.

3.) When all the pre-assigned steps are finished, the power supply will shut down the output and keep beeping to remind you until you press any key to stop the alarm. LCD will display "End" and actual running time in section 3 (Figure 4).

5. Memory Mode Operation: You can store any program in the Mn (n = 1,2…8,9) of the power supply's memory for later recall and use.

Each time you press "Start" key to begin the running, the program will be automatically saved to M0. If you need this program later, you can read it out by pressing "Read" key, "0" key and then "Enter" key. Press "Start" key to begin the running, the previous program will be carried out, again.

Note: If a new program is stored into M0, the previous contents will be erased. M0 can only store the last program that you used.

You can store a new program in the Mn (n = 1,2…8,9) of the power supply's memory for later recall and use. The operations are as follows:

⑴ Input a new group of parameters first.

⑵ Press "Save" key, "n" key (n = 1,2…8,9) and "Enter" key, then the new program will be saved to Mn. LCD will display WMn OK! (n = 1,2…8,9).

⑶ Press "Read" key, and then press "n" key (n = 1,2…8,9), LCD will display "Mode: RM" and "M = n" (n = 1,2…8,9).

⑷ Press "Enter" key, then the program stored in Mn (n = 1,2…8,9) appears. LCD will display "RMn OK!".

⑸ To run, simply connect the power supply with electrophoresis cell and press "Start" key.

Note: Please make sure the program is properly correct before you start the running.

Points for attention

1. If the running mode is standard mode, the output will not be shut down, when all the pre-assigned time has elapsed, but the power supply will beep 13 times to remind you.

2. If the running mode is time mode, VH mode or Step mode, the output will be shut down, when all the pre-assigned time has elapsed and the power supply will keep beeping to remind you until you press any key to stop the alarm.

3. If the power supply detect something wrong, LCD will show you "Wrong !!" and the relevant information and keep beeping to alarm you until you press any key to stop the sound of the alarm. The meaning of the information that LCD screen will display is as follows:

Ⅰ）Over Load !: The load is near to the status of short circuit;

Ⅱ）No Load ! : The load is near to the status of open circuit;

Ⅲ） Over Heat: The apparatus is in the condition of over-temperature

Ⅳ）Overrun_U: The output of the voltage is beyond the limit;

Ⅴ）Overrun_I: The output of the electric current is beyond the limit;

Ⅵ）Overrun_P: The output of the power is beyond the limit;

Ⅶ）Short !: The output terminal is short circuit;

Ⅷ）GND_Leakage! : The case of the power supply has leakage current.

If the malfunction mentioned above occurs, the s power supply will stop the output.

4. When LCD displays "No Load", turn off the power supply first, then, check the lead, the joints between the lead and the power supply, the joints between the lead and electrophoresis cell and the platinum wire of the electrophoresis cell.

Note: In order to avoid the unnecessary loss, the inspection should be made before you start the running.

5. If you connect the power supply with more than two units of electrophoresis cell at the same time, the output value of current that

LCD displays in section 1 (Figure 4) is the sum of all the electrophoresis cells, we suggest you select constant voltage as the running condition to reduce the interaction of different electrophoresis cells.

6. The output voltage of the power supply is very high, so please don't touch the output terminals and electrophoresis cells during the running to avoid the danger.

7. Avoid connecting a high voltage output lead to earth ground.

8. Annotation of the figures on the front panel:

Ⅰ） Red "+" on the top right corner of output terminals: Anode output terminals;

Ⅱ） Black "-" on the top left corner of output terminals: Cathode output terminals;

Ⅲ） The figure of "ϟ" in yellow triangle: Indicates high voltages may be present;

Ⅳ） The figure of "!" in yellow triangle: Attention! Please refer to the documents that we supply with the power supply.

Care and maintenance

- You should check if the lead or power cord is still good after a certain period to avoid some unexpected malfunctions.
- You should often wipe off the dust and dirt on the surface of the electrophoresis power supply with a soft cloth wetted with clean water or a mild cleaning solution.
- Turn off the apparatus and pull the power cord from the wall outlet if you do not use it for a period of time, and the apparatus should be covered by protection casing.
- Don't store the apparatus in moist environment, the products after being packed should be stored in (－20 ～55) ℃ / (-4 ～131) °F, and relative humidity does not exceed 93% , the store room should not has corrosive gas and should be breezy.

Replacing fuse

If there is no display in LCD screen after you turn on the power supply, the fuse may need to be replaced. And the operation steps are as follows:

- Turn off the power supply and pull the power cord from the wall outlet. **Note:** Please make sure to hold the plug attachment and not pull the power cord directly, and please don't hold the plug by wet hand.
- Unscrew the screw-cap of the fuse socket on the back side of the power supply in counterclockwise direction, and take out the fuse tube, you can judge the fuse tube by visual determination or by Ohm classification of universal meter, and replace it with a new fuse tube.
- Screw the screw-cap in clockwise direction.

Technical service and repair

We offer complete technical support for all of our products, if you have any questions about how to use this apparatus or if you find that we can help you, please feel free to contact us by call or fax or E- mail. If you would like to arrange to repair it, please remember to tell us the following information including the model, serial number, purchase date, etc.

Ordering information

Description	Quantity	Cat. Number
Electrophoresis Power Supply (Model: DYY － 15D)	1	113 – 1540
Power cord	1	326 – 0001
User's Manual	1	---